Scheduled Test Questions

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Q No. 1

Question

The area of the triangle formed by the line

$$\frac{x}{3} + \frac{y}{4} = 1$$

with the coordinate axes is:

A) 3

B) 6

C) 4

D) 12

Correct Answer

	6

Solution

	NA

Q No. 2

Question

The slope of the line passing through $(2, 3)$ and $(5, 9)$ is:

A) 2

B) 1

C) 3

D) 4

Correct Answer

	2

Solution

	NA

Q No. 3

Question

Find the maximum value of $3\sin x + 4\cos x$.

A) 4

B) 7

C) 5	
D) 12	
Correct Answer	
	5
Solution	
	NA

Q No. 4	
Question	
The solutions of $\sin x = \sin 50^\circ$ in the interval $0^\circ \leq x \leq 360^\circ$ are:	
A) $50^\circ, 100^\circ$	
B) $50^\circ, 230^\circ$	
C) $50^\circ, 130^\circ$	
D) $130^\circ, 310^\circ$	
Correct Answer	
	$50^\circ, 130^\circ$
Solution	
	NA

Q No. 5	
Question	
If $\sec\theta + \tan\theta = 3$, then find $\tan\theta$.	
A) $\frac{2}{3}$	
B) $\frac{4}{3}$	
C) $\frac{3}{4}$	
D) $\frac{5}{3}$	
Correct Answer	
	$\frac{4}{3}$
Solution	

	NA

Q No. 6

Question

Simplify:

$$\frac{1+\tan^2\theta}{1+\cot^2\theta}$$

A) $\sin^2\theta$

B) $\tan^2\theta$

C) $\cos^2\theta$

D) $\cot^2\theta$

Correct Answer

	$\tan^2\theta$

Solution

	NA

Q No. 7

Question

If $\tan\theta+\cot\theta=5$, then find $\tan^2\theta+\cot^2\theta$.

A) 21

B) 25

C) 23

D) 27

Correct Answer

	23

Solution

	NA

Q No. 8

Question

If $\sin\theta+\cos\theta=\sqrt{2}$, then find $\sin 2\theta$.

A) $\frac{1}{2}$

B) 0

C) 1

D) -1

Correct Answer

	1

Solution

	NA

Q No. 9

Question

If $\sin\theta < 0$ and $\tan\theta > 0$, then θ lies in:

A) First quadrant

B) Third quadrant

C) Second quadrant

D) Fourth quadrant

Correct Answer

	Third quadrant

Solution

	NA

Q No. 10

Question

If $\tan\theta = \frac{3}{4}$, where θ is acute, then find $\sin\theta + \cos\theta$.

A) $\frac{1}{5}$

B) $\frac{5}{7}$

C) $\frac{7}{5}$

D) $\frac{3}{5}$

Correct Answer

	$\frac{7}{5}$

Solution	
	NA

Q No. 11	
Question	
Find the maximum value of $7\sin x + 24\cos x$.	
A) 24	
B) 25	
C) 31	
D) 49	
Correct Answer	
	25
Solution	
	NA

Q No. 12	
Question	
If $\sin\theta + \cos\theta = 1$, then find $\sin\theta\cos\theta$.	
A) 0	
B) $\frac{1}{2}$	
C) 1	
D) -1	
Correct Answer	
	0
Solution	
	NA

Q No. 13	
Question	
Find the value of $\sin 90^\circ - \cos 180^\circ$.	
A) 0	
B) 1	
C) 2	

D) -2

Correct Answer

	2

Solution

	NA

Q No. 14

Question

If $\tan\theta = 1$, where $0° < \theta < 90°$, then θ is:

A) $30°$

B) $45°$

C) $60°$

D) $90°$

Correct Answer

	$45°$

Solution

	NA

Q No. 15

Question

Find the value of $\sec 60° + \ cosec 30°$.

A) 2

B) 3

C) 4

D) 5

Correct Answer

	4

Solution

	NA

Q No. 16

Question

Simplify $1 - \sin^2\theta$.

A) $\sin^2\theta$

B) $\cos^2\theta$

C) $\tan^2\theta$

D) $\sec^2\theta$

Correct Answer

	$\cos^2\theta$

Solution

	NA

Q No. 17

Question

If θ lies in the third quadrant, then which of the following is positive?

A) $\sin\theta$

B) $\cos\theta$

C) $\tan\theta$

D) $\sec\theta$

Correct Answer

	$\tan\theta$

Solution

	NA

Q No. 18

Question

If $\cos\theta = \frac{12}{13}$, where θ is acute, then find $\tan\theta$.

A) $\frac{5}{12}$

B) $\frac{12}{5}$

C) $\frac{5}{13}$

D) $\frac{13}{5}$

Correct Answer

	$\frac{5}{12}$

Solution

	NA

Q No. 19

Question

Find the value of $\frac{\cos 30^\circ}{\sin 60^\circ}$.

A) 0

B) $\frac{1}{2}$

C) 1

D) $\sqrt{3}$

Correct Answer

	1

Solution

	NA

Q No. 20

Question

Convert 210° into radians.

A) $\frac{5\pi}{6}$

B) $\frac{7\pi}{6}$

C) $\frac{4\pi}{3}$

D) $\frac{3\pi}{2}$

Correct Answer

	$\frac{7\pi}{6}$

Solution

	NA

Q No. 21

Question

If $\tan A + \cot A = 7$, find the value of $\tan^2 A + \cot^2 A$.

Correct Answer

	47

Solution

	NA

Q No. 22

Question

If $\sin\theta = \frac{5}{13}$, where θ is acute, find the value of $13(\sin\theta + \cos\theta)$.

Correct Answer

	17

Solution

	NA

Q No. 23

Question

Find the value of $8\cos 0° + 6\sin 90° - 5\tan 45°$.

Correct Answer

	9

Solution

	NA

Q No. 24

Question

Find the value of:

$$4\sin 30° + 6\cos 60° + 3\tan 45°$$

Correct Answer

	8

Solution

	NA

Q No. 25

Question

Find the distance between the points $(5, 7)$ and $(-3, 1)$

Correct Answer

	10

Solution

	NA

Q No. 26

Question

The ratio of masses of O_2 and N_2 in a particular gaseous mixture is 1:4.
The ratio of number of their molecules is

A) 1:8

B) 3:16

C) 1:4

D) 7:32

Correct Answer

	7:32

Solution

	$\frac{n_{O_2}}{n_{N_2}} = \frac{\frac{1}{32}}{\frac{4}{28}} = \frac{7}{32}$

Q No. 27

Question

Select the correct option for decreasing order of number of atoms.
(I) 2 gm of hydrogen
(II) 16 gm of sulphur
(III) 4 gm of oxygen
(IV) 31 gm of phosphorus

A) II > III > IV > I

B) IV > II > III > I

C) I > IV > II > III

D) I = IV > II > III

Correct Answer

	I > IV > II > III

Solution

(I) 2 gm of hydrogen

number of moles $= \frac{2}{1} = 2 \times N_A$ number of atoms

(II) 16 gm of sulphur

number of moles $= \frac{16}{32} = \frac{1}{2} \times N_A$ number of atoms

(III) 4 gm of oxygen

number of moles $= \frac{4}{16} = \frac{1}{4} \times N_A$ number of atom

(IV) 31 gm of phosphorus

number of moles $= \frac{31}{31} = 1 \times N_A$ number of atoms

$I > IV > II > III$

Q No. 28

Question

Which among the following has lowest number of atoms?

A) 10 gm $CaCO_3$

B) 2.24 litre of CO_2 at STP

C) 2 moles of H_2 gas

D) 5 gm-atom of Na_2CO_3

Correct Answer

2.24 litre of CO_2 at STP

Solution

Number of atoms in $10 gm CaCO_3$

$= \frac{10}{100} \times 6.02 \times 10^{23} \times 5 = 3.01 \times 10^{23}$ atoms

Number of atoms in 2. 24 litre of CO_2

$= \frac{2.24}{22.4} \times 6.02 \times 10^{23} \times 3 = 1.86 \times 10^{23}$ atoms

Number of atoms in 2 mole of H_2

$= 2 \times 6.02 \times 10^{23} \times 2 = 12.08 \times 10^{23}$ atoms

Number of atoms in 5 gm -atom of Na_2CO_3

$= 5 \times 6.02 \times 10^{23} \times 6 = 18.07 \times 10^{24}$ atoms

Q No. 29

Question

In which case is the number of molecules of water maximum?

A) 18 mL of water

B) 10^{-3} mol of water

C) 0.00224 L of water vapours at 1 atm and 273 K

D) 0.18 g of water

Correct Answer

18 mL of water

Solution

* 10^{-3} mole water $= 6.02 \times 10^{20}$ molecule H_2O

* 18 ml H_2O = 18 gram = 1 mole

$= 6.02 \times 10^{23}$ molecules

* At 1 atm & 273 K, no. of moles of H_2O

$= \frac{0.00224}{22.4} = \frac{22.4 \times 10^{-4}}{22.4}$

$= 10^{-4}$ moles $= 6.02 \times 10^{19}$ molecules

* 0.18 gram $H_2O = 0.1$ moles $= 6.02 \times 10^{22}$ molecules

Q No. 30

Question

22 gm of ideal gas occupies 11.2 L volume at 1 atm & 273 K. The gas(es) may be

A) SO_2

B) N_2O

C) NO_2

D) CO

Correct Answer

N_2O

Solution

$V = 11.2$ L; $P = 1$ atm; $T = 273$ K

$\therefore n = \frac{11.2}{22.4} = \frac{1}{2}$ mol

wt. = 22 gm and $n = \frac{\text{wt.}}{\text{Mol wt.}}$

$\therefore$ M. wt. $= \frac{\text{wt.}}{n} = \frac{22}{\frac{1}{2}} = 22 \times 2 = 44$ gm

M. wt. of $N_2O = 44$ gm

Q No. 31

Question

1.0 g of H_2 has same number of molecules as in:

A) 14 g of N_2

B) $18\ \text{g of } H_2O$

C) $16\ \text{g of } CO$

D) $28\ \text{g of } N_2$

Correct Answer

$14\ \text{g of } N_2$

Solution

$$\text{Number of moles of } H_2 = \frac{1}{2} = 0.5$$

$$\text{Number of molecules of } H_2 = 0.5\ N_A.$$

$$\text{(A) Number of moles of } N_2 = \frac{14}{28} = 0.5$$

$$\text{Number of molecules of } N_2 = 0.5\ N_A$$

$$\text{(B) Number of moles of } H_2O = \frac{18}{18} = 1$$

$$\text{Number of molecules of } H_2O = 1 \times N_A = N_A$$

$$\text{(C) Number of moles of } CO = \frac{16}{28} = \frac{4}{7}$$

$$\text{Number of molecules of } CO = \frac{4}{7}\ N_A$$

$$\text{(D) Number of moles of } N_2 = \frac{28}{28} = 1$$

$$\text{Number of molecules of } N_2 = 1 \times N_A = N_A$$

Q No. 32

Question

If N_A is Avogadro number, then the number of valence electrons in $4.2\ \text{g of } N^{-3}$ ions is

A) $2.4\ N_A$

B) $4.2\ N_A$

C) $1.6\ N_A$

D) $3.2\ N_A$

Correct Answer

$2.4\ N_A$

Solution

Moles of $N^{-3} = \frac{4.2}{14} = 0.3$

One ion N^{-3} contain 8 valence electrons

$\therefore$ Moles of valence e^- in $4.2\ g$ of $N^{-3} = 2.4$

Q No. 33

Question

Match List I with List II

	LIST- I		**LIST- II**
A	16 g of CH_4 (g)	I	Weight 28 g
B	1 g of H_2 (g)	II	60.2×10^{23} electrons
C	1 mole of N_2 (g)	III	Weight 32 g
D	0.5 mol of SO_2 (g)	IV	Occupies 11.4 L volume at STP

Chose the correct answer from the options given below:

A) A-II, B-IV, C-I, D-III

B) A-II, B-IV, C-III, D-I

C) A-II, B-III, C-IV, D-I

D) A-I, B-III, C-II, D-IV

Correct Answer

A-II, B-IV, C-I, D-III

Solution

16g CH_4 = 1 mol CH_4 contains $10 \times 6.02 \times 10^{23}$ electrons

$= 60.2 \times 10^{23}$

1gH_2 = 0.5 mol H_2 gas occupy 11.35 lit

volume at STP {1 mol occupy 22.7 lit. volume}

1 mol N_2 (g) = 28g

0.5 mol SO_2 (g) = $0.5 \times 64 = 32$g

Q No. 34

Question

One atom of an element X weighs 6.643×10^{-23} g. Number of moles of atom in 20 kg is

A) 4

B) 40

C) 100

D) 500

Correct Answer

500

Solution

Atomic weight of an element

$$X = 6.643 \times 10^{-23} \times N_A = 40$$

No. of moles of $X = \frac{20 \times 1000}{40} = 500$

Q No. 35

Question

If Avogadro number N_A, is changed from $6.022 \times 10^{23}\ mol^{-1}$ to $6.022 \times 10^{20}\ mol^{-1}$, this would change

A) the mass of one mole of carbon.

B) the ratio of chemical species to each other in a balanced equation.

C) the ratio of elements to each other in a compound.

D) the definition of mass in units of grams.

Correct Answer

the mass of one mole of carbon.

Solution

Mass of $1\ mol\ (6.022 \times 10^{23}\ atoms)$ of carbon $= 12$

If avogadro number is changed to $6.022 \times 10^{20}\ mol^{-1}$, then mass of 1 mol. of carbon $= \frac{12 \times 6.022 \times 10^{20}}{6.022 \times 10^{23}} = 12 \times 10^{-3}\ g$

Q No. 36

Question

$2.8 \times 10^{-3} mol$ of CO_2 is left after removing 10^{21} molecules from its 'x' mg sample. The mass of CO_2 taken initially is

Given: $N_A = 6.02 \times 10^{23} mol^{-1}$

A) 48.2 mg

B) 98.3 mg

C) 150.4 mg

D) 196.2 mg

Correct Answer

196.2 mg

Solution

Le initial mass of $CO_2 = x$ mg

$\therefore$ Initial moles $= \dfrac{x \times 10^{-3}}{44}$

moles removed $= \dfrac{10^{21}}{6.02 \times 10^{23}}$

$\therefore$ moles left $= 2.8 \times 10^{-3} = \dfrac{x \times 10^{-3}}{44} - \dfrac{10^{21}}{6.02 \times 10^{23}}$

$\Rightarrow x = 196.2$ mg

Q No. 37

Question

The weight of one molecule of a compound $C_{60}H_{122}$ is

A) 1.2×10^{-20} g

B) 5.025×10^{23} g

C) 1.4×10^{-21} g

D) 6.023×10^{23} g

Correct Answer

1.4×10^{-21} g

Solution

Solution:

Molecular weight of $C_{60}H_{122} = 60 \times 12 + 1 \times 122$

$= 842$

$\therefore$ Weight of 1 mole of $C_{60}H_{122}$ molecules

$= 842$ g

$\Rightarrow$ Weight of 6.023×10^{23} molecules = 842 g

$\Rightarrow$ Weight of 1 molecule $= \dfrac{842}{6.023 \times 10^{23}}$

$= 1.4 \times 10^{-21}$ g

Q No. 38

Question

If 0.2 mole of an element X has mass $12.8\,g$, then its gram atomic mass is:

A) $32\,g$

B) $48\ g$

C) $64\ g$

D) $128\ g$

Correct Answer

$64\ g$

Solution

NA

Q No. 39

Question

In $4.4\,g$ of CO_2, the number of moles of oxygen atoms is:

A) 0.1 mole

B) 0.2 mole

C) 0.3 mole

D) 0.4 mole

Correct Answer

0.2 mole

Solution

NA

Q No. 40

Question

Which of the following contains the same number of molecules?

A) $2\,g$ H_2 and $32\,g$ O_2

B) $2\,g$ H_2 and $16\,g$ O_2

C) $4\,g$ H_2 and $32\,g$ O_2

D) $1\,g$ H_2 and $32\,g$ O_2

Correct Answer

$2\,g$ H_2 and $32\,g$ O_2

Solution

NA

Q No. 41

Question

$11.2\,g$ of iron contains:

A) 0.1 mole of Fe atoms

B) 0.2 mole of Fe atoms

C) 0.5 mole of Fe atoms

D) 1 mole of Fe atoms

Correct Answer

	0.2 mole of Fe atoms

Solution

	NA

Q No. 42

Question

Number of oxygen atoms present in $8\,g$ of O_2 is:

A) 1.505×10^{23}

B) 3.011×10^{23}

C) 6.022×10^{23}

D) 12.044×10^{23}

Correct Answer

	3.011×10^{23}

Solution

	NA

Q No. 43

Question

The mass of 3.011×10^{23} molecules of water is:

A) $4.5\,g$

B) $9\,g$

C) $18\,g$

D) $36\,g$

Correct Answer

	$9\,g$
Solution	
	NA

Q No. 44

Question

One atomic mass unit is equal to:

A) Mass of one hydrogen atom

B) Mass of one oxygen atom

C) $\frac{1}{12}$th mass of one carbon-12 atom

D) Mass of one mole of carbon atoms

Correct Answer

$\frac{1}{12}$th mass of one carbon-12 atom

Solution

NA

Q No. 45

Question

Which statement of Dalton's atomic theory was later found to be incorrect due to the discovery of isotopes?

A) Matter is made up of atoms

B) Atoms combine in simple whole-number ratios

C) Atoms of the same element are identical in all respects

D) Atoms cannot be created or destroyed in chemical reactions

Correct Answer

Atoms of the same element are identical in all respects

Solution

NA

Q No. 46

Question

The number of atoms in 8 g of sodium is $x \times 10^{23}$. The value of x is_____.(Nearest integer)

[Given : NA = 6.02×10^{23} mol^{-1}

Atomic mass of Na = 23.0 u]

Correct Answer

2

Solution

$$\text{No. of atoms} = \frac{8}{23} \times 6.02 \times 10^{23} = 2.09 \times 10^{23}$$

$$\simeq 2 \times 10^{23}$$

$$\simeq x \times 10^{23}$$

$$x = 2$$

Q No. 47

Question

Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B.

Correct Answer

2

Solution

Solution:

Given moles of A_2B = moles of AB_3

$$\frac{w}{\text{mo.wt. of } A_2B} = \frac{w}{\text{mol.wt. of } AB_3}$$

$$\frac{w}{2A+B} = \frac{w}{A+3B}$$

$$A + 3B = 2A + B$$

$$2B = A$$

So, at. Wt. of A is 2 times that of B

Q No. 48

Question

Find the mass of $CaCO_3$ containing 3.011×10^{22} formula units.

Correct Answer

	5

Solution

	NA

Q No. 49

Question

A metal oxide has formula M_2O_3. If 0.10 mole of this oxide weighs $16\,g$, find the atomic mass of metal M.

Correct Answer

	56

Solution

	NA

Q No. 50

Question

Find the mass of 0.75 mole of $Al_2(SO_4)_3$.

Correct Answer

	255,257

Solution

	NA

Q No. 51

Question

The set builder form of the set A = {1, 15, 53, 127, 249, …….} is

A) $A = \{x : x = 4n^2 - 1; n \in W\}$

B) $A = \{x : x = n + 14; n \in N\}$

C) $A = \{x : x = n^3 - 1; n \in N\}$

D) $A = \{x : x = 2n^3 - 1; n \in N\}$

Correct Answer

$A = \{x : x = 2n^3 - 1; n \in N\}$

Solution

Option (4) is correct as putting n = 1, 2, 3, 4, 5, …….

$2n^3 - 1$ takes values 1, 15, 53, 127, 249, ……

Q No. 52

Question

Let $n(U) = 700, n(A) = 200, n(B) = 300$ and $n(A \cap B) = 100$, then $n(A^c \cap B^c) =$

A) 400

B) 600

C) 300

D) 200

Correct Answer

300

Solution

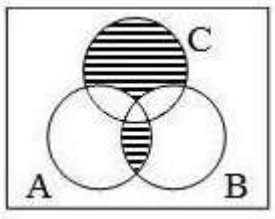
$$n(A^C \cap B^C) = n[(A \cap B)^c] = n(U) - n(A \cup B)$$
$$= n(U) - [n(A) + n(B) - n(A \cap B)]$$
$$= 700 - [200 + 300 - 100] = 300$$

Q No. 53

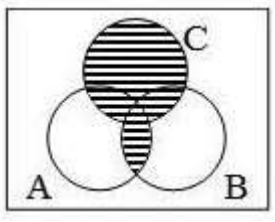
Question

$(A \cap B \cap C') \cup (A' \cap B' \cap C)$ is represented by

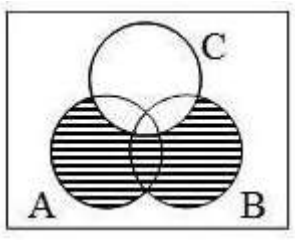
A)



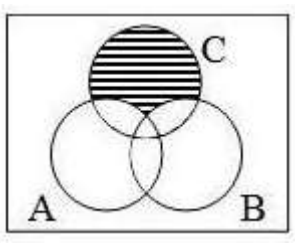
B)



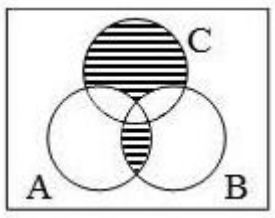
C)



D)

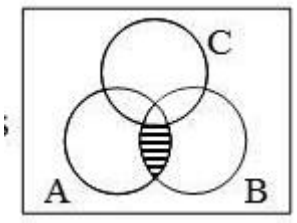


Correct Answer



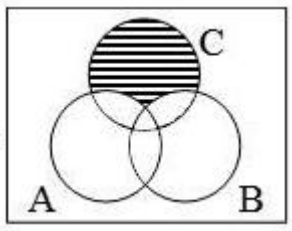
Solution

$\because (A \cap B \cap C')$ is

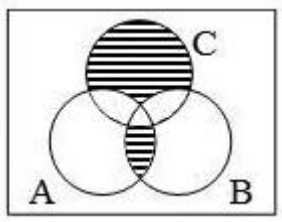


and

$(A' \cap B' \cap C)$ is



Therefore $(A \cap B \cap C') \cup (A' \cap B' \cap C)$ is



Q No. 54

Question

25 people are employed for programme A, 50 people are employed for programme B, 10 people for both. So, number of employees employed only for A is

A) 15

B) 20

C) 35

D) 40

Correct Answer

15

Solution

$$n(A - B) = n(A) - n(A \cap B)$$
$$= 25 - 10 = 15$$

Q No. 55

Question

Which of the following is a singleton set?

A) $\{x : x^2 + 1 = 0;\ x \in R\}$

B) $\{x : x^2 - 1 = 0;\ x \in R\}$

C) $\{x : x^3 + 2x^2 + 2x + 1 = 0;\ x \in R\}$

D) $\{x : x^3 - 6x^2 + 11x - 6 = 0;\ x \in R\}$

Correct Answer

$\{x : x^3 + 2x^2 + 2x + 1 = 0;\ x \in R\}$

Solution

The equation $x^3 + 2x^2 + 2x + 1 = 0 \Rightarrow (x+1)(x^2 + x + 1) = 0$ is satisfied by only real x which is equal to $'-1'$

Q No. 56

Question

If number of elements in A is 3 and number of elements in power set of B is 24 more than the number of elements in power set of A then number of elements in B is

A) 5

B) 4

C) 3

D) 2

Correct Answer

5

Solution

$n(\mathrm{A}) = 3$
$\Rightarrow n(P(\mathrm{A})) = 2^3 = 8$
let $n(\mathrm{B}) = n$ then,
$2^n = 8 + 24 = 32 = 2^5$ [given]
$\Rightarrow n = 5$

Q No. 57

Question

Two finite sets have m and n elements. The number of subsets of the first set is 112 more than that of the second set. The values of m and n are, respectively

A) 4, 7

B) 7, 4

C) 4, 4

D) 7, 7

Correct Answer

7, 4

Solution

Since, number of subsets of a set containing *m* elements is 122 more than the subsets of the set containing *n* elements.

$\because 2^m - 2^n = 122$

$\Rightarrow 2^n \cdot (2^{m-n} - 1) = 2^4 \cdot 7$

$\Rightarrow 2^n = 2^4$ and $2^{m-n} - 1 = 7$

$\Rightarrow n = 4$ and $2^{m-n} = 8$

$\Rightarrow 2^{m-n} = 2^3 \Rightarrow m - n = 3$

$\Rightarrow m - 4 = 3 \Rightarrow m = 4 + 3$

$\Rightarrow m = 7$

Q No. 58

Question

The number of non-empty subsets of the set {1, 2, 3, 4} is

A) 15

B) 14

C) 16

D) 17

Correct Answer

15

Solution

Solution:

The number of non-empty subsets = $2^n - 1$

$2^4 - 1 = 16 - 1 = 16$

Q No. 59

Question

Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$

A) 3

B) 6

C) 9

D) 18

Correct Answer

6

Solution

Solution:

$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 3 + 6 - n(A \cap B)$

Since, maximum number of elements in $A \cap B = 3$

$\therefore$ Minimum number of elements in $A \cup B = 9 - 3 = 6$

Q No. 60

Question

In a class of 100 students, 55 students have passed in Mathematics and 67 students have passed in Physics. Then the number of students who have passed in Physics only is

A) 22

B) 33

C) 10

D) 45

Correct Answer

45

Solution

Solution:

$n(M) = 55, n(P) = 67, n(M \cup P) = 100$

Now, $n(M \cup P) = n(M) + n(P) - n(M \cap P)$

$100 = 55 + 67 - n(M \cap P)$

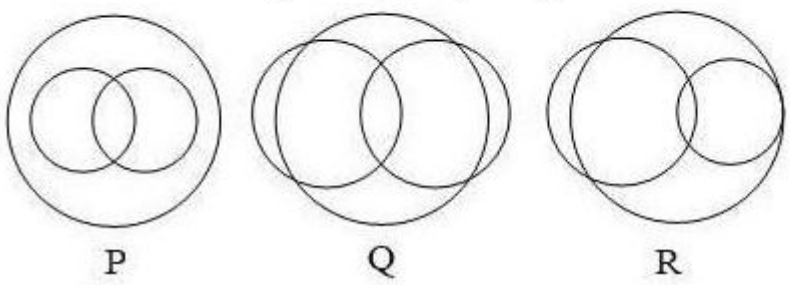
$\therefore n(M \cap P) = 122 - 100 = 22$

Now $n(P \text{ only}) = n(P) - n(M \cap P) = 67 - 22 = 45$

Q No. 61

Question

In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement?



A) P and Q

B) P and R

C) None of these

D) Q and R

Correct Answer

None of these

Solution

$A \cap B \cap C$ is visible in all three venn diagram. So, none of these.

Q No. 62

Question

Given $n(U)=20, n(A)=12, n(B)=9, n(A \cap B)=4$, where U is the universal set, A and B are subsets of U, then $n((A \cup B)^c)=$

A) 17

B) 9

C) 11

D) 3

Correct Answer

3

Solution

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$= 12 + 9 - 4 = 17$$

Now,

$$n((A \cup B)^c) = n(U) - n(A \cup B)$$
$$= 20 - 17 = 3$$

Q No. 63

Question

If A and B are two sets then $(A - B) \cup (B - A) \cup (A \cap B)$ is equal to

A) $A \cup B$

B) $A \cap B$

C) A

D) B'

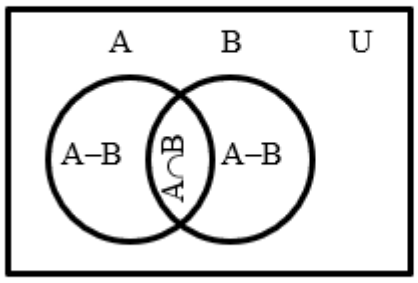
Correct Answer

$A \cup B$

Solution

Solution:

Form vein diagram,



$\therefore (A - B) \cup (B - A) \cup (A \cap B) = A \cup B$

Q No. 64

Question

Given $A = \{x \mid x \text{ is a root of } x^2 - 1 = 0\}$, $B = \{x \mid x \text{ is a root of } x^2 - 2x + 1 = 0\}$, then

A) $A \cap B = A$

B) $A \cup B = \phi$

C) $A \cup B = A$

D) $A \cap B = \phi$

Correct Answer

$A \cup B = A$

Solution

Solution:

We have, $A = \{X \mid x \text{ is a root of } x^2 - 1 = 0\}$

$= \{x \mid x \text{ is a root of } (x - 1)(x + 1) = 0\} \Rightarrow x = \pm 1$

$B = \{x \mid x \text{ is root of } (x - 1)^2 = 0\} \Rightarrow x = 1$

$\Rightarrow A \cup B = A.$

Q No. 65

Question

If $A = \{x : x = 6^n - 5n - 1; n \in N\}$

and $B = \{x : x = 25(n - 1); n \in N\}$

Then $A \cap B$ is

A) $A - B$

B) $B - A$

C) A

D) B

Correct Answer

A

Solution

$A = \{0, 25, 200, 1275, \ldots\}$ and $B = \{0, 25, 50, 75, \ldots\}$

Clearly $A \subset B \Rightarrow A \cap B = A$

Q No. 66

Question

Consider the following relations

(1) $A - B = A - (A \cap B)$

(2) $A = (A \cap B) \cup (A - B)$

(3) $A - (B \cup C) = (A - B) \cup (A - C)$

Which of these is/are correct

A) 1 and 3

B) 2 only

C) 2 and 3

D) 1 and 2

Correct Answer

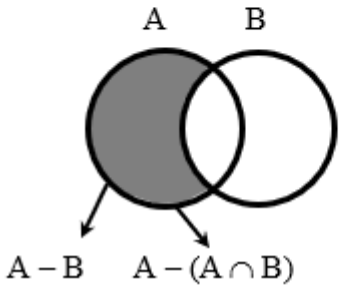
1 and 2

Solution

Solution:

$A - B = A - (A \cap B)$ is correct.

$A = (A \cap B) \cup (A - B)$ is correct.



Q No. 67

Question

If $n(A)=8$ and $n(A\cap B)=2$, then $n[(A\cap B)'\cap A]$ is equal to

A) 2

B) 4

C) 6

D) 8

Correct Answer

6

Solution

$n[(A\cap B)'\cap A]$ [By De Morgan's law]
$=n[(A'\cup B')\cap A]$ [Distributive law]
$=n(A-B)=8-2=6$

Q No. 68

Question

If S = {x | x is a positive multiple of 3 less than 100} and P = {x | x is a prime number less than 20}. Then, n(S) + n(P) is equal to

A) 34

B) 31

C) 33

D) 41

Correct Answer

41

Solution

Solution:

$\because$ S = {x | x is a positive multiple of 3 less than 100}

$\therefore$ n(S) = 33 and P = {x | x is a prime number less than 20}.

$\therefore$ n(P) = 8

n(S) + n(P) = 33 + 8 = 41.

Q No. 69

Question

Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point P(m, n) from the point Q (–2, –3) is:

A) 6

B) 4

C) 10

D) 8

Correct Answer

10

Solution

As given, $2^m - 2^n = 56$

$\Rightarrow$ m = 6, n = 3

P(6, 3) and Q(–2, –3)

Distance PQ $= \sqrt{(6+2)^2 + (3+3)^2}$
$=10$

Q No. 70

Question

The number of elements in the set $\{(a, b): 2a^2 + 3b^2 = 35, a, b \in Z\}$, where Z is the set of all integers, is

A) 2

B) 4

C) 8

D) 12

Correct Answer

8

Solution

Given set is $\{(a, b): 2a^2 + 3b^2 = 35, a, b, \in Z\}$
We can see that, $2(\pm 2)^2 + 3(\pm 3)^2 = 35$
And $2(\pm 4)^2 + 3(\pm 1)^2 = 35$
$\therefore (2, 3), (-2, -3), (-2, 3), (4, 1), (4, -1), (-4, -1), (-4, 1)$ are 8 elements of the set.
$\therefore n = 8$

Q No. 71

Question

If number of elements in A is 3 and number of elements in power set of B is 24 more than the number of elements in power set of A then number of elements in B is

Correct Answer

5

Solution

$n(A) = 3 \Rightarrow n(P(A)) = 2^3 = 8$

let $n(B) = n$ then, $2^n = 8 + 24 = 32 = 2^5$ [given]

$\Rightarrow n = 5$

Q No. 72

Question

If $A = \{1,2,3,4,5\}, B = \{2,4,6\}, C = \{3,4,6\}$, Then $n((A \cup B) \cap C)$ is

Correct Answer

3

Solution

$A \cup B = \{1,2,3,4,5,6\}$
$\therefore (A \cup B) \cap C = \{3,4,6\}$

Q No. 73

Question

If A = {x : x is a multiple of 4} and B = {x: x is a multiple of 6} then $A \cap B$ consists of all multiples of

Correct Answer

12

Solution

$A = \{4, 8, 12, 16, 20, 24,.....\}; B = \{6, 12, 18, 24, 30,.....\}$

$A \cap B = \{12, 24,.....\} = \{x : x \text{ is multiple of } 12\}$

Q No. 74

Question

In a town of 840 persons, 450 persons read Hindi, 300 red English and 200 read both. The number of persons who read neither, is

Correct Answer

290

Solution

Let H be the set of person who read Hindi and E be the set of persons who read English. Then,

$n(U) = 840, n(H) = 450, n(E) = 300, n(H \cap E) = 200$

Number of persons who read neither = $n(H' \cap F')$

$\Rightarrow n(H \cup E)'$

$\Rightarrow n(U) - n(H \cup E)$

$= 840 - [n(H) + n(E) - n(H \cap E)]$

$= 840 - (450 + 300 - 200) = 840 - 550 = 290$

Q No. 75

Question

Let $X = \{x : 1 \leq x \leq 50, x \in N\}$

$A = \{x: x \text{ is multiple of } 2\}$

$B = \{x: x \text{ is multiple of } 7\}$

Then find number of elements in the smallest subset of X which contain elements of both A and B

Correct Answer

29

Solution

Solution:

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 25 + 7 - 3$

$= 29$